

Narendran L

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GitHub: github.com/Narendran-ds | Portfolio: narendranl-portfolio.vercel.app

Summary

Software engineer specializing in generative AI, with hands-on experience building and deploying LLM-powered pipelines: a Gemini API system generating 500+ cuisines at Syncorb Geotech, and ChainSight, an explainable AI application pairing a fine-tuned YOLOv8 model with Gemini-based prompt engineering for natural-language narration. Also shipped ZipForgeX, a Spring Boot and React platform with OAuth2 and PostgreSQL. Comfortable across dataset prep, model integration, prompt optimization, and systematic testing (58 automated tests), while working fluently with AI coding tools (Claude, ChatGPT).

Skills

Languages: Python, Java, SQL

Generative AI: Gemini API, LLM Integration, Prompt Engineering, PyTorch, PaddleOCR, YOLOv8 Fine-Tuning

Data & Analytics: pandas, NumPy, scikit-learn, XGBoost, SHAP, Power BI, Excel

AI-Assisted Development: Claude, ChatGPT (debugging, prototyping, accelerated iteration)

Backend & Deployment: Spring Boot, REST APIs, PostgreSQL, Git, GitHub, Render, Vercel, Streamlit

Experience

AI Developer Intern

Oct 2025 - May 2026

Syncorb Geotech Pvt Ltd, Chennai

- Designed and built a Gemini API pipeline to generate 500+ global cuisines and up to 500 dishes per cuisine, replacing a manual BeautifulSoup/Scrapy workflow with prompt-engineered LLM calls; added a history-tracking system so completed entries are skipped on reruns, cutting redundant API usage.
- Rebuilt the restaurant menu extraction pipeline from scratch using PaddleOCR after third-party API access was restricted, implementing folder-based batch processing and history tracking to replace an incomplete prototype with a production-ready system.
- Worked directly with pre-trained models and third-party AI APIs end-to-end: prompt design, output validation, error handling, and pipeline reliability.

Projects

ChainSight — Python, YOLOv8, NetworkX, Shapely, Gemini, Streamlit

Code: github.com/Narendran-ds/chainsight

- Architected a five-layer perception pipeline (Vision → Spatial → World Graph → Rule Engine → Narration) that converts raw warehouse video into auditable safety events, deployed live on Streamlit Cloud with 58 passing tests.
- Trained a YOLOv8 detector on a merged 17-class, 13,731-image dataset, reaching mAP@50 of 0.77 (0.89 AP@50 on exit-zone markers) after leakage-safe three-way splitting.
- Built a deterministic rule engine (R1–R5) over a NetworkX world graph with Shapely/STRtree spatial reasoning, emitting events with explicit trigger, evidence, and conclusion; constrained a Gemini Flash narration layer to rephrase-only outputs to prevent hallucinated facts.

ZipForgeX — Spring Boot, Java 17, React, PostgreSQL, NVIDIA NIM, OAuth2

Live at zipforgex.in

- Built and deployed a full-stack app (Spring Boot 3.5 backend, React + Vite + Tailwind frontend) that converts indented folder-structure text into downloadable ZIP archives, live on Render + Vercel under a custom domain.
- Integrated an AI folder-structure generator through a JWT-gated `/api/ai/generate-structure` endpoint calling NVIDIA NIM, with a system prompt locked to 4-space indentation for parser compatibility.
- Implemented JWT + Spring Security auth with Google and GitHub OAuth2, plus ZIP history and account management on PostgreSQL/Supabase, and hardened the ZIP pipeline against Zip Slip path-traversal attacks – covered by 12 passing tests.

Education

B.Tech in Artificial Intelligence and Data Science

2023 - 2027 (Expected)

Rajalakshmi Institute of Technology, Poonamallee, Chennai

CGPA: 7.53/10

Class 12

Completed March 2023

Velammal Vidyalaya, Paruthipattu

Achievements and Leadership

- Shortlisted from Rajalakshmi Institute of Technology to represent the college at Cognizant Technoverse Hackathon 2026, a national-level hackathon with 22,000+ participants.
- Scout and Guide Leader: Led teams and demonstrated discipline in school-level camps.
- Discipline Committee Leader: Faculty-selected peer conduct manager.